LIBRARY MANAGEMENT SYSTEM

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Abstract

The Library Management System is a simple desktop application developed using Java Swing, JDBC, and MySQL. It provides an interface to maintain book records and perform basic database operations.

Introduction

Managing book details manually can be time-consuming. This application stores book information in a MySQL database and allows users to manage records through a graphical interface.

Objectives

- To connect a Java application to a MySQL database using JDBC.
- To add, view, update, delete, and search book records.
- To store records permanently in a database and verify database operations.

Technologies Used

Java Swing, JDBC, MySQL, MySQL Workbench, and Apache NetBeans.

Database Design

Database: library_db
Table: books
Fields: book_id (primary key), title, author, category, and quantity.

Main Modules

- Add Book: inserts new book details into MySQL.
- View Books: displays saved book records in a table.
- Update Book: modifies the selected book record.
- Delete Book: removes the selected record after confirmation.
- Search Book: searches records by title or author.

Database Operations

The application uses INSERT to add books, SELECT to retrieve records, UPDATE to edit details, and DELETE to remove records. Input validation checks that required fields are entered and quantity is a non-negative whole number.

Screenshots

The following screenshots show the application interface, MySQL table, successful insertion, and the NetBeans project/build.

Figure 1. Library Management System application interface

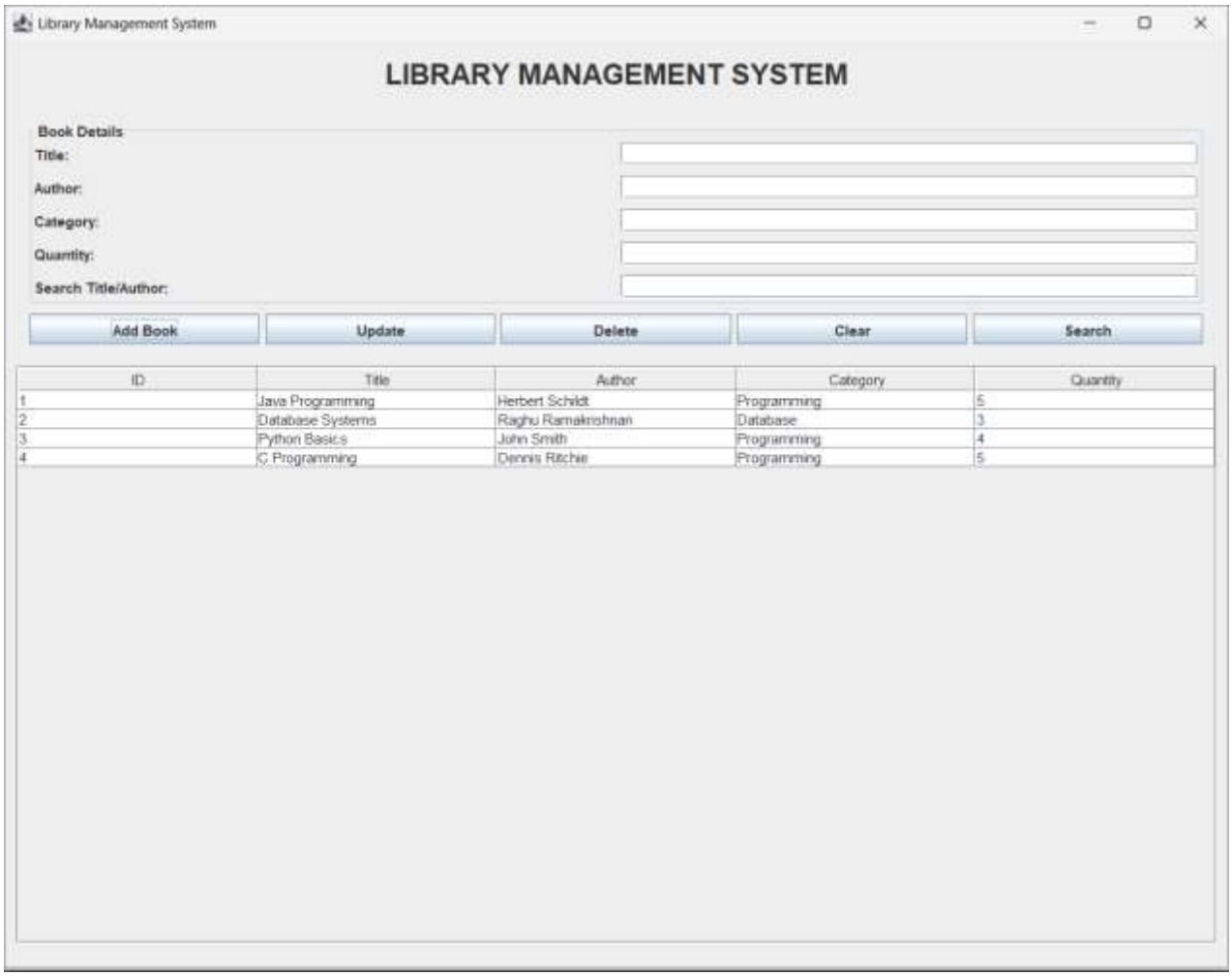


Figure 2. MySQL Workbench database and books table

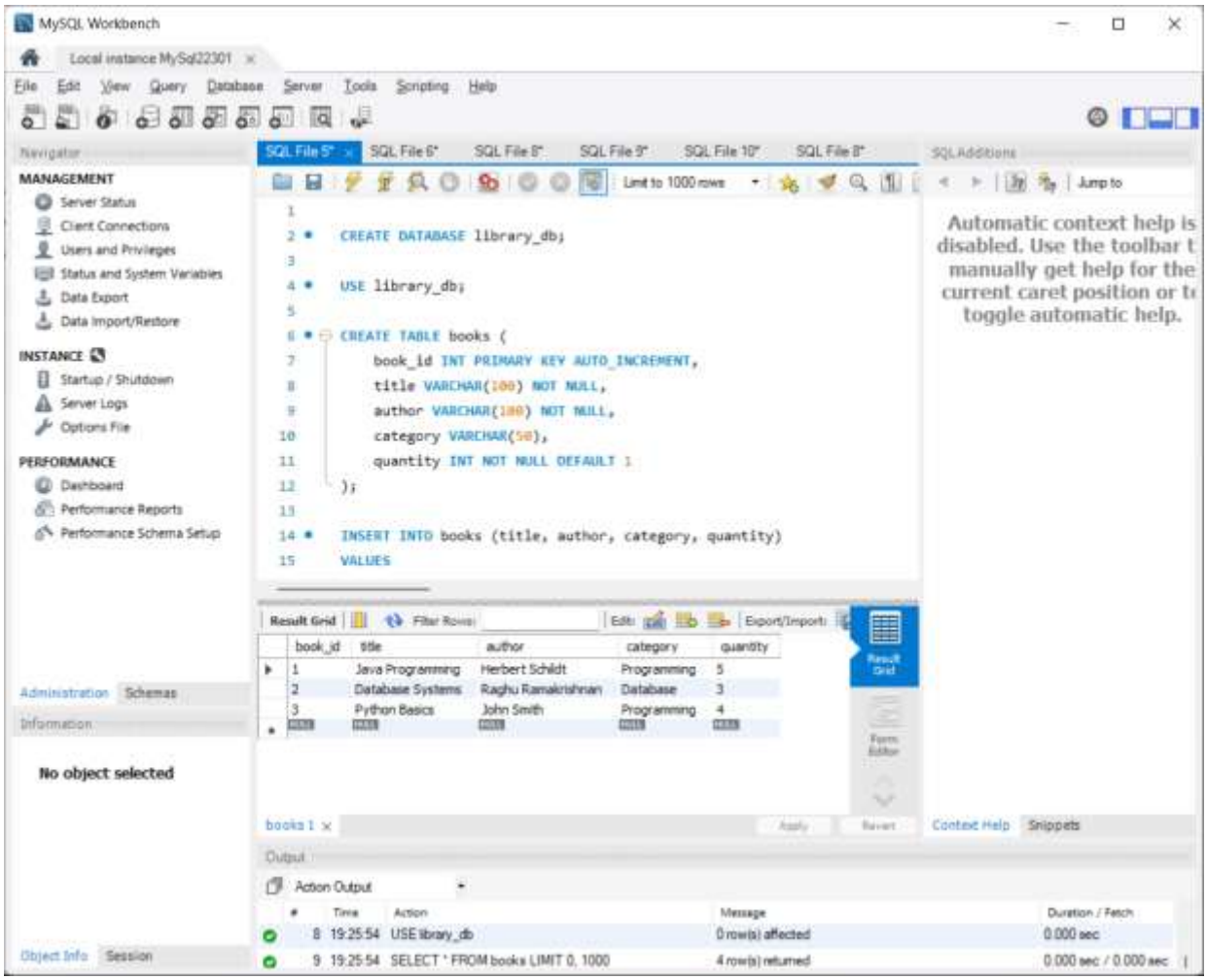


Figure 3. Book inserted successfully through the application

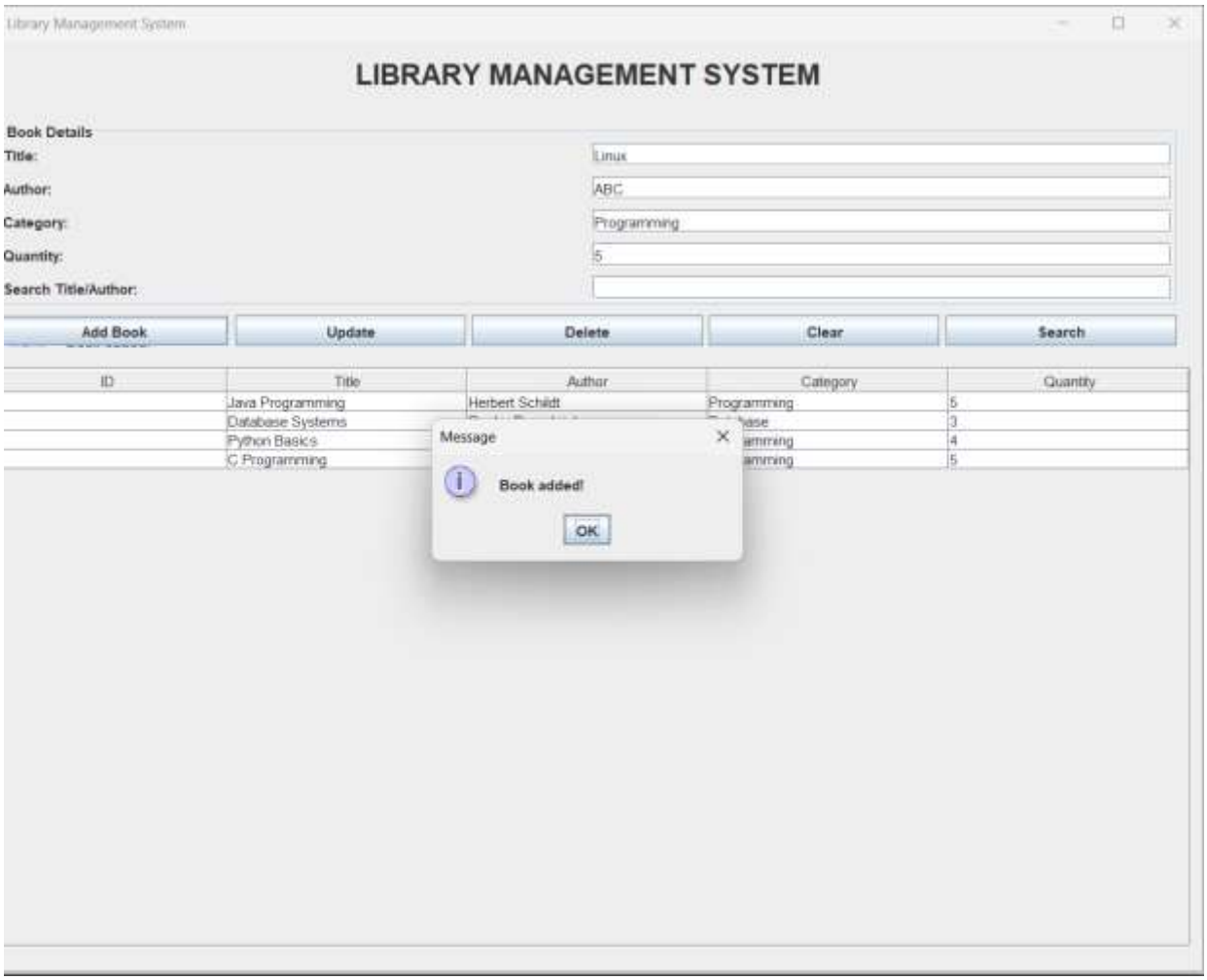
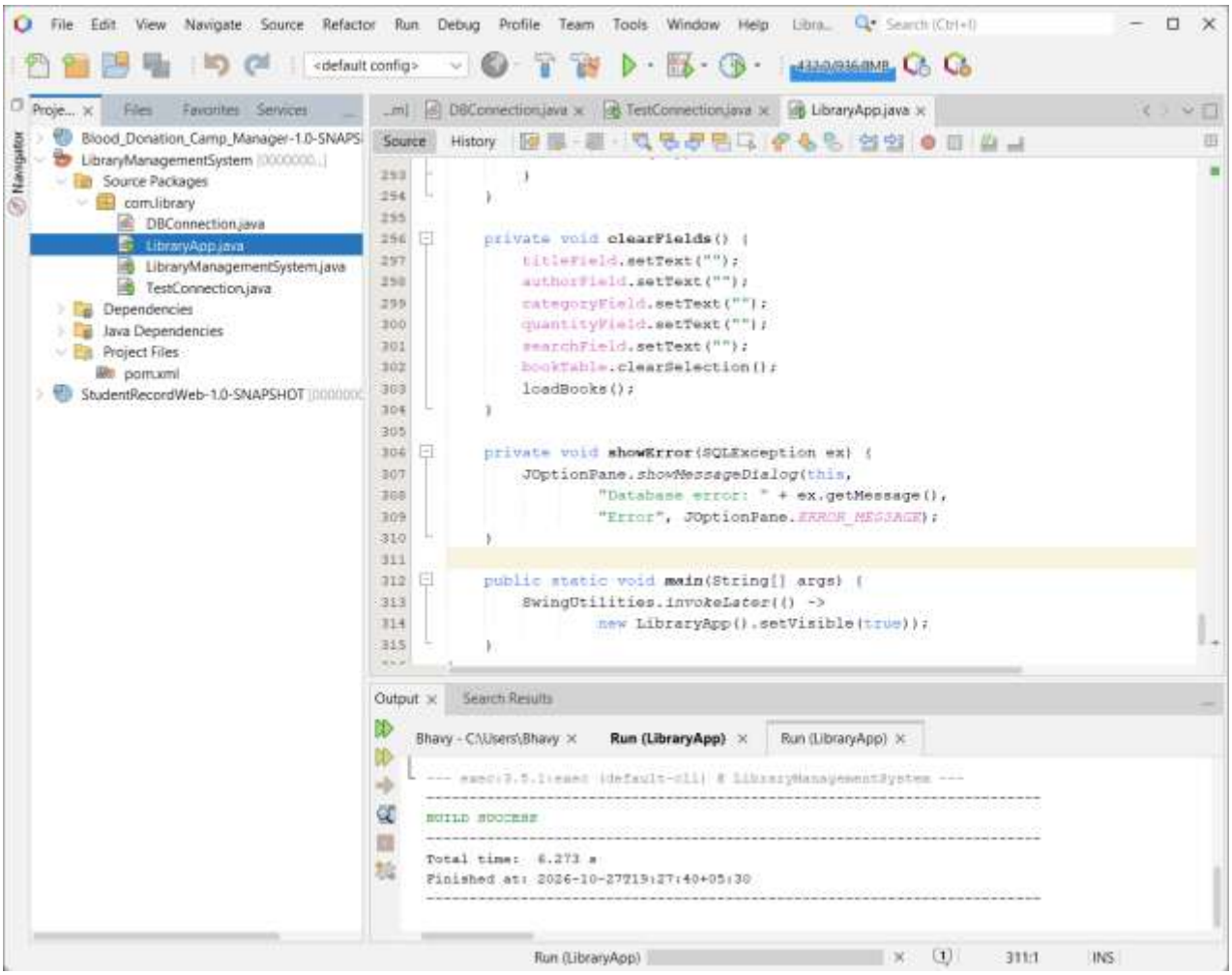


Figure 4. NetBeans project structure and successful build



Testing and Results

The application opened and displayed records from MySQL. A new book was entered through the application and a success message was displayed. The database can be checked in MySQL Workbench using SELECT * FROM books;. Update, delete, search, and validation should be verified before final demonstration.

Conclusion

The project demonstrates how Java Swing can be connected to MySQL using JDBC. It provides a simple way to maintain book records and demonstrates the core CRUD database operations.

Future Enhancements

Future improvements could include librarian login, issue/return tracking, due-date reminders, and reports for available books.